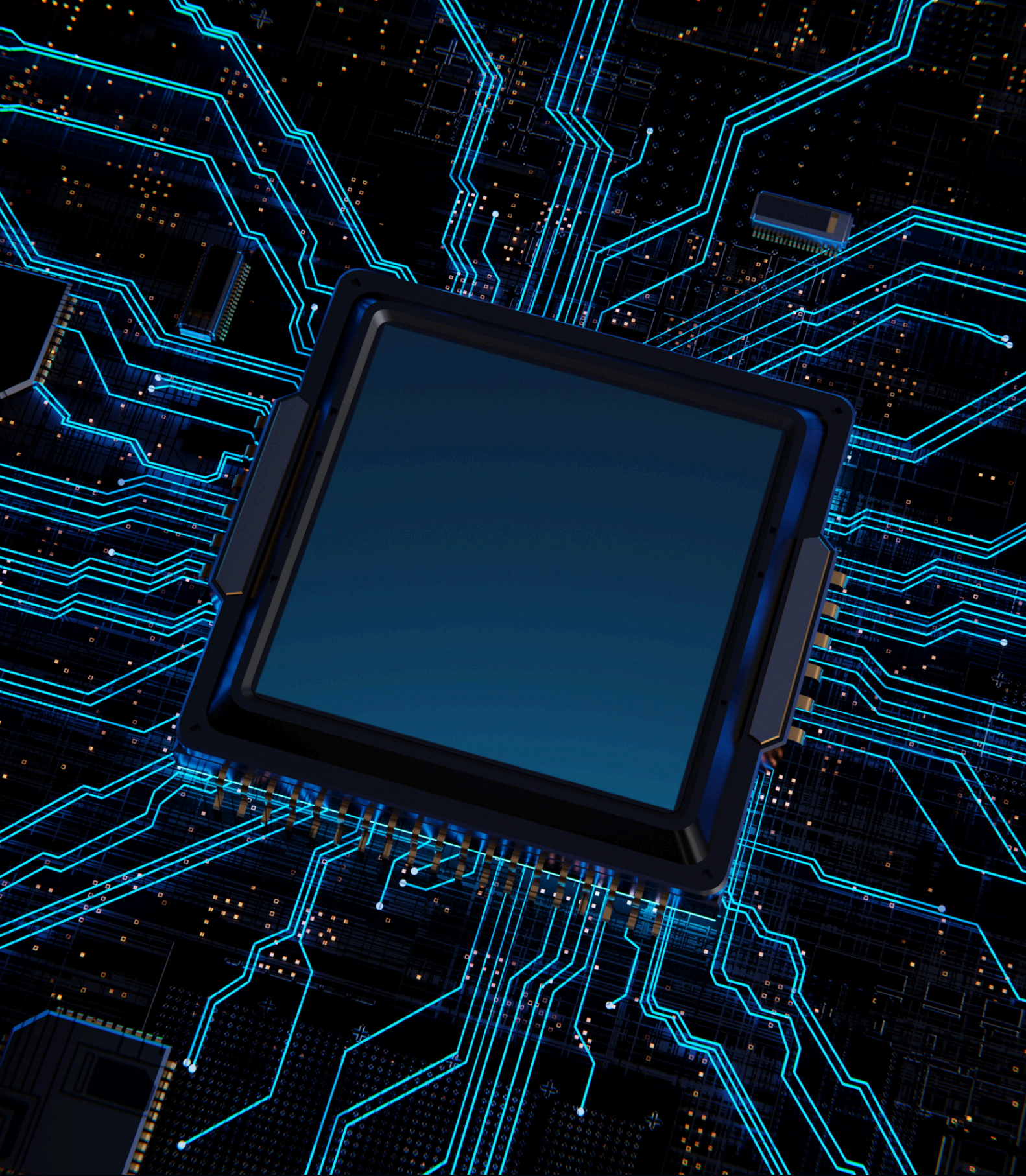




PHISHING DETECTION & AWARENESS REPORT

SOUMALYA JANA
CS | TASK 02



1. Introduction

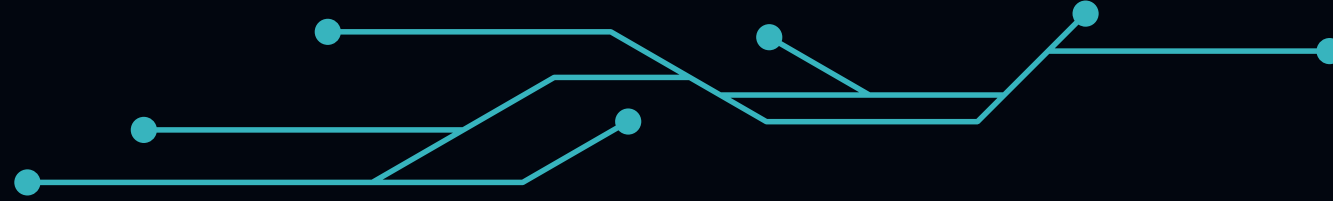
Phishing is a common cybersecurity attack where attackers attempt to trick individuals into revealing sensitive information such as passwords, banking details, or personal data.



2. Objective

The objective of this project is to:

- Identify phishing characteristics in email samples
- Classify emails as Safe, Suspicious, or Phishing
- Explain phishing techniques in simple, user-friendly language
- Provide awareness and prevention guidelines for users.



3. METHODOLOGY

THE FOLLOWING APPROACH WAS USED TO ANALYZE THE EMAIL SAMPLES:

- **VERIFYING THE SENDER ' S EMAIL ADDRESS FOR AUTHENTICITY**
- **EXAMINING SUBJECT LINES FOR URGENCY OR THREATENING LANGUAGE**
- **CHECKING EMBEDDED LINKS FOR SUSPICIOUS OR FAKE DOMAINS**
- **REVIEWING EMAIL CONTENT FOR TONE, GRAMMAR, AND INTENT**
- **IDENTIFYING REQUESTS FOR SENSITIVE INFORMATION OR ATTACHMENTS**



Email Analysis

Email 1

Subject: Important: Verify Your Bank Account

Sender: security@bank-update.com

Classification: Phishing

Indicators:

- Fake domain not associated with a legitimate bank
- Request for sensitive financial information
- Threatening tone (account closure warning)
- Mention of attachments that may contain malware

Explanation:

This email is designed to steal banking information by pressuring the user. Attackers use fear tactics to make users act quickly without verifying the authenticity of the message.

Email Analysis

Email 2

Subject: Important: Verify Your Bank Account

Sender: security@bank-update.com

Classification: Phishing

Indicators:

- Fake domain not associated with a legitimate bank
- Request for sensitive financial information
- Threatening tone (account closure warning)
- Mention of attachments that may contain malware

Explanation:

This email is designed to steal banking information by pressuring the user. Attackers use fear tactics to make users act quickly without verifying the authenticity of the message.

5. COMMON PHISHING TECHNIQUES

Attackers commonly use the following techniques:

- Email Spoofing: Creating fake sender addresses that appear legitimate
- Urgency and Fear Tactics: Pressuring users to act immediately
- Fake Websites: Redirecting users to malicious websites that mimic real ones
- Link Manipulation: Hiding malicious URLs behind trusted-looking text
- Attachment-Based Attacks: Sending infected files to install malware



6. PREVENTION & AWARENESS GUIDELINES

Effects

To protect against phishing attacks, users should follow these best practices:

- Always verify the sender ' s email address carefully
- Avoid clicking on suspicious or unknown links
- Hover over links to check the actual URL before clicking
- Do not download unexpected attachments
- Never share sensitive information via email
- Enable Two-Factor Authentication (2FA) Use updated antivirus and security software
- Report suspicious emails to the appropriate authority or IT team